NAKITOMA SEED SECONDARY SCHOOL

END OF TERM I 2026

SENIOR THREE CHEMISTRY ASSESSMENT

TIME: 2 HOURS

INSTRUCTIONS:

- ***Attempt all items***
- ***Responses to item one(1) and item two(2) should be written in the spaces provided***
- ***Responses to item three(3) and item four(4) should be written on the answer sheets provided***

SECTION A

ITEM 1

Mr. Ssebulime a farmer in Kakumiro has for two years suffered with a challenge of low crop yields. He of recent invited senior agricultural officials from the district who examined the current situation of his soil. They gave him a report that his soil has turned acidic due to over use of organic manure. In order to solve this problem, he was advised to apply slaked lime (calcium hydroxide) to his soil. This left him confused and he has approached you for advice.

a) Help Mr. Ssebulime understand the category of slaked lime.

.....

.....

b) Briefly describe how slaked lime can solve Mr. Ssebulime's challenge.

.....

.....

.....

.....

c) Identify any one side effect of excessive use of slaked lime to the soil with its mitigation measure.

.....

.....

.....

d) Give at least one difference and one similarity of acids and the category of substances identified in a) above.

I. Similarity.

.....

.....

II. Difference.

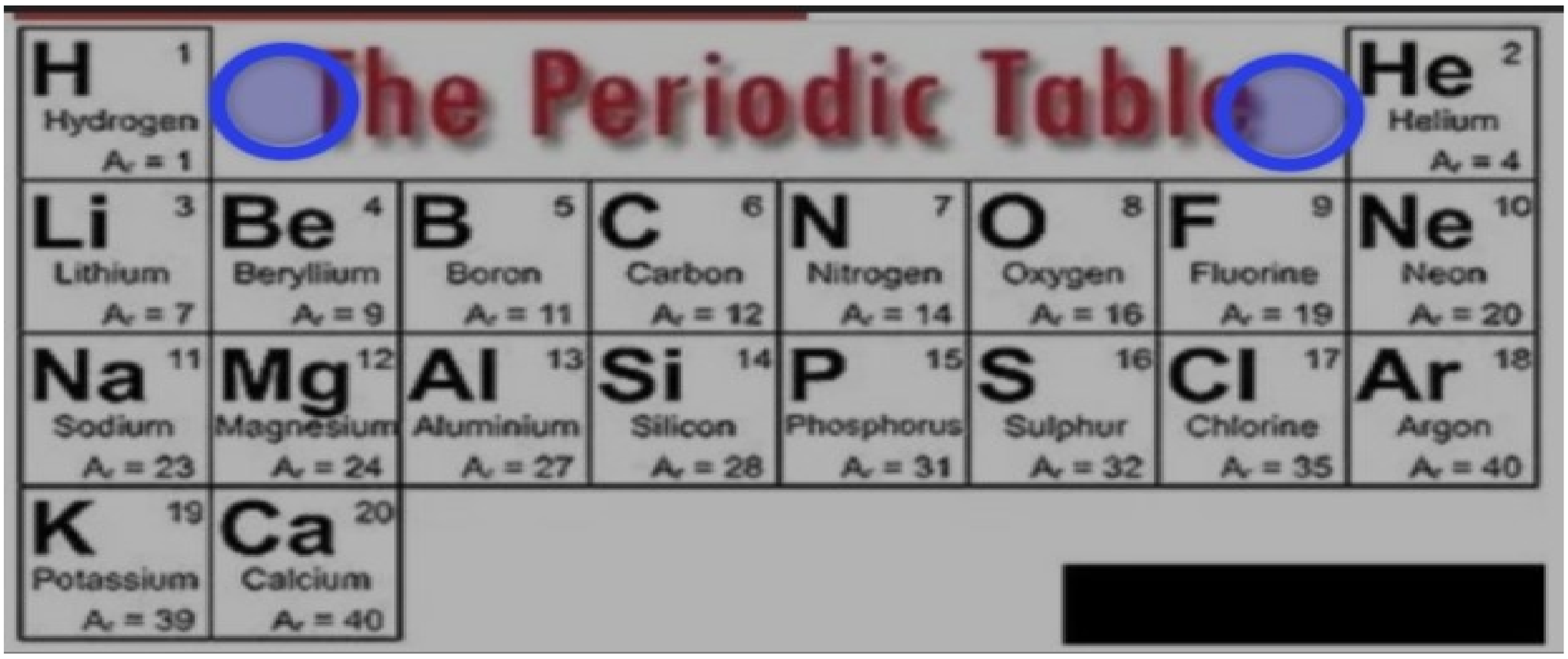
.....

.....

.....

ITEM 2

Nambooze a 50 year old, complains of dental caries, joint pains and muscular crumps. During one of her visits to a hospital, the doctor advised her to drink water from borehole because it contains a compound that can solve her problems. Nambooze would like to know more about the compound and has approached you for assistance. Your chemistry teacher told you that the compound is chemically written as XY_2 (where X and Y are not actual symbols of the elements). The teacher also informed you that X belongs to group II and period 4 while Y belongs to group VII and period 3 of the periodic table.



TASK

a. Explain the nature of substances X and Y and identify them.

.....

.....

.....

..

b. Explain how X and Y form a compound.

.....

.....

.....

.....

c. Explain the properties of the compound formed between X and Y.

.....

.....

.....

.....

d. Write down any one impact of having a lot of compound XY_2 in the environment and its mitigation.

.....

.....

.....

.....

.....

SECTION B

PART 1

ITEM 3

Nitrogen is one of the essential components used in the manufacture of ammonia which is used in the manufacture of ammonium fertilizers. Uganda being a country which depends on Agriculture, there is a high demand for nitrogen to be used for the above processes. The government has therefore decided to set up a nitrogen manufacturing plant in Buseesa village in Bugweri district, however the local people are not contented as they would like to first know how the production process will be, its side effects and social benefits and you have been approached for advice in one of their meetings.

TASK

As a chemistry learner, make a write up you will use to address them.

PART II

ITEM 4

Air is an important natural resource that should be conserved. There is improper waste management in some homes and schools involving burning of garbage, waste plastic bottles and open dumping. These activities release toxic fumes into the atmosphere hence affecting the quality of air. The head teacher has tasked the science club to organize a sensitization campaign aimed at solving the challenge. The theme of the campaign is **"Be an air quality champion"**. You have been chosen to speak to the rest of the students regarding the theme.

TASK

Make a write up of the message you will present. (Include the category of the natural resource, its chemical composition, how human activity is affecting it and its benefits if conserved)

END